Toroidal Manifolds and Recursive Neural Dynamics: Grid Cell Geometry, Stability, and Information Persistence

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November 5, 2026

Abstract

A toroidal pattern in grid cell activity raises two questions. Does nearby activity return to the torus after a displacement, and which distinctions can an observer still recover? We audit five recording files and analyze a model of two coupled oscillators. The archived population matrix combines cell labels across trials, so its topology cannot answer the recovery question. In the model, two circular phases give the same toroidal state space across several regimes. Independent rotations can be periodic or dense, and coupling can lock the relative phase onto a circle. Adding two amplitude coordinates lets us change transverse stability while leaving every trajectory on the torus unchanged. A perturbation away from the torus distinguishes these models by its measured return rate. The same contraction preserves an exactly observed binary label for every finite time, although a finite resolution readout eventually gives both labels the same value. These examples specify what a neural account of continuity would have to establish about the state, its recovery, and the observer's resolution.

1 Introduction

Grid cells provide a tractable setting for studying how neural populations represent continuous variables. Individual cells fire at locations arranged approximately on a triangular lattice [1]. Across a population, relative firing patterns can encode position within that repeating structure. This periodic representation motivates toroidal coordinates and raises a dynamical question: what maintains the representation when neural activity is perturbed?

Published recordings support both geometric and dynamical descriptions of grid cell populations. Gardner et al. recovered toroidal population topology within simultaneously recorded modules, including across waking and sleep conditions [2]. Hermansen et al. found toroidal representations during one dimensional behavior [3]. Yoon et al. showed that relationships among simultaneous cells remained conserved despite changes in individual responses, supporting a low dimensional continuous attractor [4]. These findings motivate models in which nearby activity returns toward a continuous family of represented states [5].

The relation between spatial tuning and population dynamics nevertheless requires separate measurements. Di Sarra et al. found that timing perturbations could disrupt recovered topology while leaving spatial grid scores largely intact [6]. Their independent rate modulated Poisson model also produced toroidal signatures; a later correction replaces duplicated panels in the synthetic calibration figure [7]. A toroidal representation can thus be supported by several constructions, each with different implications for the underlying dynamics.

Two coupled oscillators allow a direct test of what population geometry can reveal about dynamics. Neural rhythms and cross frequency interactions motivate phase descriptions [8–10]. Two independently specifiable phases define a toroidal state space. Their equations determine where a trajectory goes on that torus. To test attraction, one must also specify what happens to a state displaced away from it.

Our central construction gives two systems with identical trajectories on the torus but opposite responses to a transverse displacement. No observation confined to the torus can distinguish the two systems through those trajectories alone. A second construction addresses information persistence: two initially distinct states contract toward the same limit, and the observer's resolution determines how long their difference can be read. Together these constructions give the observer a precise role in a theory of continuity.

Section 2 specifies the recordings, models, and observations. Section 3 compares the archived measurements with the locking, recovery, and information calculations. Section 4 describes a neural recovery experiment and its relevance to accounts of conscious experience. The derivations and reproducibility details follow in Appendices A–H.

2 Methods

2.1 Recording Audit and Observables

We examine five files from the Hafting grid cell recordings [1, 11]. The data documentation identifies simultaneously recorded cells within each file. We treat each file separately unless acquisition metadata establishes that two files share a recording session. For the archive audit, we reconstruct the older procedure that placed all files on one time axis. We compare its output entry by entry with the stored population matrix.

The spatial analysis assigns each observed map bin a probability proportional to its firing rate. This yields a categorical entropy of spatial tuning. We also inspect the dimensions and lifetimes in the retained persistence diagrams. Appendix C gives the recording inventory, normalization, and preprocessing settings.

2.2 Oscillator Model

The model represents each oscillator by a phase $\phi_i \in \mathbb{R}/2\pi\mathbb{Z}$ with constant natural frequency ω_i. Their joint coordinate space is $T^2 = S^1 \times S^1$. We use symmetric sinusoidal

coupling [12, 13],

$$\begin{aligned} \dot{\phi}_1 &= \omega_1 + \epsilon \sin(\phi_2 - \phi_1), \\ \dot{\phi}_2 &= \omega_2 + \epsilon \sin(\phi_1 - \phi_2), \qquad \epsilon \geq 0. \end{aligned} \tag{1}$$

Here $\epsilon = 0$ gives independent rotations. With phase difference $\delta = \phi_2 - \phi_1$ and detuning $\Delta\omega = \omega_2 - \omega_1$, the relative dynamics reduce to

$$\dot{\delta} = \Delta\omega - 2\epsilon \sin \delta. \tag{2}$$

The lifted common phase advances at $(\omega_1 + \omega_2)/2$. This reduction identifies synchronization without assuming attraction transverse to the torus.

Normal attraction is tested by adding positive amplitude coordinates r_1, r_2 with

$$\dot{r}_i = -\lambda(r_i - 1), \qquad \lambda > 0. \tag{3}$$

The full state is embedded as $(r_1 \cos\phi_1, r_1 \sin\phi_1, r_2 \cos\phi_2, r_2 \sin\phi_2)$ in $\mathbb{R}^4$. This extension specifies the directions in which states can approach the unit product torus. Independent rotations, phase locking, radial perturbations, and a solvable forcing perturbation are compared under the same model framework. Appendices A and B provide the derivations and stability conditions.

2.3 Numerical and Observation Checks

Finite simulations test the analytic predictions in arbitrary model units. The principal frequencies are $(1, \sqrt{2})$, with coupling strengths 0, 0.1, and 0.6; a rational frequency control uses $(1, 3/2)$. Fourth order Runge–Kutta trajectories are compared with exact solutions and adaptive integration. Phase kicks and radial displacements test recovery in different directions. Step refinement measures error over a fixed interval; Appendix D specifies all parameters, tolerances, and convergence limits.

Information is evaluated using specified observation maps. An exactly encoded binary variable and its finite resolution observation test whether contraction preserves distin-

guishability. Projections of the coupled phase model test whether an observed variable has autonomous dynamics. These examples use constructed states and observation rules; their definitions and calculations are in Appendix E.

3 Results

3.1 What the Retained Recordings Establish

The archived population matrix combines trials that took place at different times. The five files contain 23 cell recordings under 13 distinct labels. Merging spikes with repeated labels on one recording's time axis reproduces all entries in the stored matrix of 59,992 time bins by 13 labels. A row in that matrix therefore has no single time at which the represented population was observed together.

The retained diagrams contain H_1 intervals from embeddings of this pooled matrix. The repository contains no matching H_2 output. The archived spatial maps answer a different question: across 13 maps the mean entropy of the normalized rates is 10.860 bits, compared with 11.288 bits for uniform rates across all 2,500 bins. This number describes how firing rates are distributed across space. Appendix C details the computation and identifies the additional topology reports that could not be reproduced.

3.2 Synchronization and Normal Attraction

The same toroidal phase coordinates support different orbit structures. With independent nonzero frequencies, a rational frequency ratio gives periodic orbits and an irrational ratio gives dense quasiperiodic orbits. Both phase directions remain neutral under perturbations. Under positive coupling, stable locking occurs when $|\Delta\omega| < 2\epsilon$, with

$$\delta_* = \arcsin\left(\frac{\Delta\omega}{2\epsilon}\right), \qquad \lambda_\delta = -\sqrt{4\epsilon^2 - (\Delta\omega)^2} < 0. \tag{4}$$

The stable locked set is a circle of fixed phase difference. It is a periodic orbit when the common frequency is nonzero and a circle of equilibria otherwise. The common phase

remains neutral. At $|\Delta\omega| = 2\epsilon > 0$ the locking exponent vanishes. When $|\Delta\omega| > 2\epsilon$, the relative phase drifts. The complete branch and threshold analysis is given in Appendix A.

The amplitude extension establishes attraction to an invariant torus by an exact distance law. For $M = \{r_1 = r_2 = 1\}$,

$$d(z(t), M) = e^{-\lambda t} d(z(0), M). \tag{5}$$

For the same initial phases on M, the choices λ, 0, and $-\lambda$ give identical phase histories. A small transverse displacement decays, stays fixed, or grows, respectively, as long as the radial coordinates remain positive. Under phase locking, M remains invariant and normally attracting even though typical limiting motion lies on a circle within it. A sufficiently strong normal contraction also gives persistence under small autonomous perturbations, subject to normal hyperbolicity [14]. This persistence concerns the manifold; preserving quasiperiodic internal motion requires further hypotheses [15].

The finite calculations agree with the predicted locking, drift, and recovery. For $(\omega_1, \omega_2) = (1, \sqrt{2})$ and $\epsilon = 0.6$, the stable phase difference is approximately 0.3524 rad and its exponent is -1.1262 per model time unit. At $\epsilon = 0.1$, the mean slip rate is 0.3627 rad per model time unit. Phase kicks persist for independent rotations, whereas locking restores relative phase and retains a common offset. The radial examples recover the predicted normal contraction. Fixed-interval refinement gives approximately fourth order convergence before the numerical floor. These results verify the finite model calculations; they supply no new biological measurements. Figure 2 and the error analysis are in Appendix D.

3.3 Information Persistence and Coarse Dynamics

Normal contraction does not by itself determine the recoverability of an encoded variable. In the binary example, two distinct initial displacements remain distinguishable at every finite time under exact observation. A fixed finite resolution eventually maps both to the same observed bin. Information loss thus depends on the encoding and measurement,

even when the dynamical contraction is unchanged. Differential entropy in the normal coordinates decreases under contraction, but that change does not specify a thermodynamic entropy flux.

An observation supports autonomous coarse dynamics only if its evolution is consistent across states with the same observation. For fine and coarse flows F_t, G_t, a continuous surjection π, and a time conversion $c > 0$, the required relation is

$$\pi \circ F_t = G_{ct} \circ \pi. \tag{6}$$

The single phase ϕ_1 fails this test when coupling is nonzero: its velocity depends on the unobserved phase. The circular sum $(\phi_1 + \phi_2) \pmod{2\pi}$ instead follows an exact autonomous rotation. Appendix E derives the consistency condition and gives both examples. Continuity across descriptions therefore requires a compatible map and dynamics.

4 Discussion

4.1 From Population Geometry to Neural Dynamics

The results identify the extra evidence needed to connect a toroidal representation with a neural attractor. Published population studies support geometric organization and, in the case of Yoon et al., a dynamical attractor interpretation for their recordings. The present audit limits what can be inferred from this repository's retained outputs. The published studies use their own simultaneously recorded populations. In our amplitude model, even complete phase trajectories on the torus leave normal stability undetermined.

A recovery experiment would estimate the manifold from simultaneous recordings of an identified module, then hold that estimate fixed when evaluating displaced states in separate trials. Let $d(t)$ denote the distance from a displaced population state to that manifold in a specified activity metric. For $d(t_1), d(t_2) > 0$ and $t_2 > t_1$, the model predicts

$$\frac{\log[d(t_1)/d(t_2)]}{t_2 - t_1} = \lambda. \tag{7}$$

The sign distinguishes contraction, neutrality, and local repulsion in the model. A phase shift along the torus can persist throughout the return. In a recording, one would compare the decay with matched control trials, account for sensory and behavioral inputs, and estimate uncertainty above the distance imposed by measurement noise. Appendix C discusses the separate calibration required to identify topology.

The oscillator results also distinguish synchronization from a two dimensional attracting representation. Coupling can reduce typical limiting motion to a circle while leaving the coordinate torus intact. An attracting torus in a larger space may coexist with this internal locking. A neural interpretation therefore needs a specified state space, a recovery measurement, and a description of the motion that remains on the recovered set.

4.2 Entropy, Observation, and Continuity

Information persistence requires both dynamics and an observation rule. Exact smooth contraction can preserve distinguishable states, whereas finite resolution can erase their observed difference. A report of decreasing entropy must accordingly identify its variable, normalization, and units. Spatial Shannon entropy, differential entropy, description length, and thermodynamic entropy describe different quantities. The thermodynamic balance in Appendix E requires a physical boundary and measured exchanges before it can constrain neural energetic cost.

Recursive processing inherits the distinctions preserved by its inputs. Repeating a deterministic operation on the same observation cannot recover distinctions already removed by that observation. New inputs can supply additional information. Predictive coding and Minimum Description Length motivate conditional compression models [16–18], but do not establish monotonic compression through every biological layer. Likewise, Eq. (6) gives a condition to test when relating scales; oscillation or shared topology alone does not satisfy it.

For a theory of subjective continuity, the open question is which differences in neural state remain available to a later report or action. The binary example shows why that

question requires a specified readout and interval. A test in humans would have to define the reported distinction, identify the neural observable used to predict it, and compare predictions at the observer's available resolution. The rat recordings considered here concern spatial coding. They contain no report of experience. Appendix F considers operator order, and Appendix G specifies what a physical model would need to connect different frequency ranges.

5 Conclusion

Two phase coordinates determine the topology of the model's state space. Coupling determines whether relative phase drifts or locks, while the amplitude equations determine whether a displaced state returns to the torus. The archived grid cell maps describe spatial tuning. Testing transverse recovery requires a population observed together through time.

The observer enters through the distinctions that a readout preserves. In the model a binary distinction survives exact contraction at every finite time, yet its two readouts eventually coincide at fixed resolution. A claim of continuity across scales must identify that readout and show when fine states that look alike have the same observable evolution. These are measurable conditions for connecting neural dynamics to the persistence of a representation.

A Geometry and Phase Dynamics

A.1 Periodic Spatial Coordinates

An idealized grid code has toroidal coordinates because positions repeat under lattice translations. Let $\Lambda = \mathbb{Z}\mathbf{e}_1 + \mathbb{Z}\mathbf{e}_2$ be a rank two lattice. Equivalent positions differ by an element of Λ, and the quotient $\mathbb{R}^2/\Lambda$ is a torus. Writing $\mathbf{x} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ gives circular coordinates $\phi_i = 2\pi u_i \pmod{2\pi}$. Figure 1 illustrates this construction. The lattice coefficients are wrapped; Cartesian coordinates are not generally independent periods of the same length.

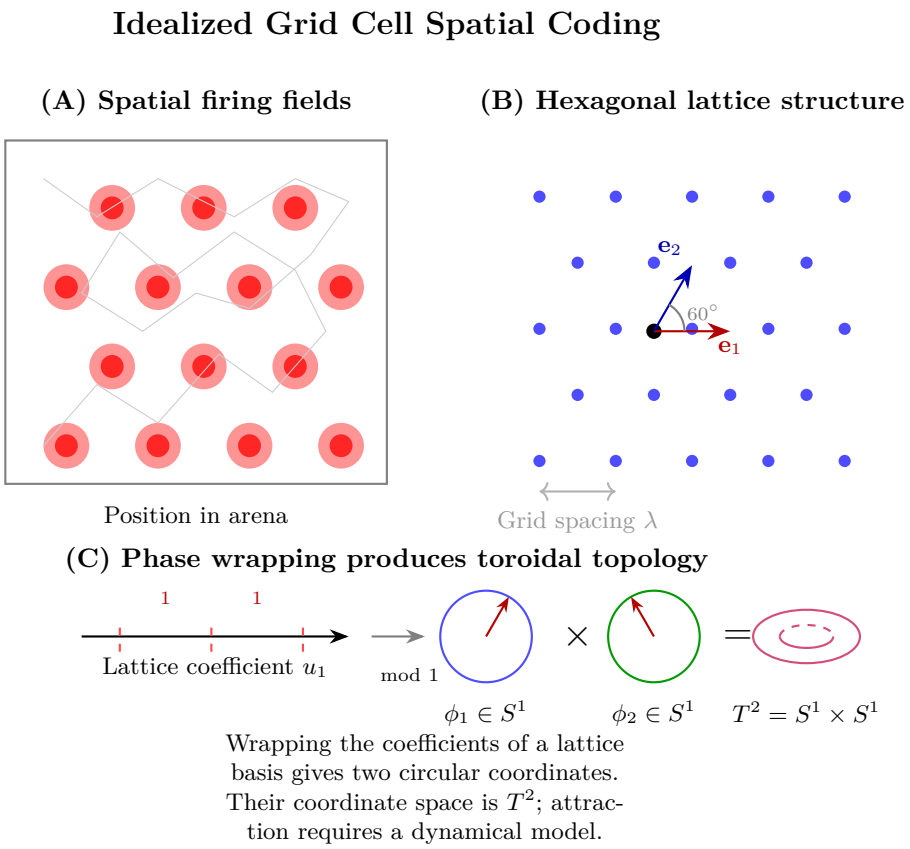


Figure 1: Idealized periodic spatial coding. (A) Schematic firing fields, not measured rate maps. (B) Two lattice basis vectors separated by 60°. (C) Wrapping their coefficients modulo one gives two circular coordinates and the quotient T^2.

A population firing map requires further assumptions to represent this quotient faithfully. It must separate distinct lattice phases and sample them adequately. A scalar firing rate from one neuron need not distinguish all phases. Spatial periodicity therefore

defines candidate coordinates without establishing an embedded population manifold or its dynamical stability.

Even a torus-compatible Betti signature does not uniquely determine a general space. Over a field, T^2 has $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$. The wedge $S^1 \vee S^1 \vee S^2$ has the same ranks. Additional assumptions or invariants are needed to identify the space. An unresolved H_2 feature is inconclusive about the underlying topology and says nothing by itself about attraction.

A.2 Circular Coordinates

A phase coordinate records position within an oscillation cycle [19, 20]. Consider a single idealized oscillator with phase variable $\theta_i(t)$ evolving as

$$\theta_i(t) = \omega_i t + \theta_{i,0}, \tag{A.1}$$

where $\theta_i(t)$ is the instantaneous phase of the i-th oscillator, ω_i is its angular frequency, and $\theta_{i,0}$ is the initial phase. The phase is identified modulo 2π, so $e^{i\theta_i(t)}$ lies on S^1. For $\omega_i \neq 0$ it traverses that circle periodically; $\omega_i = 0$ gives a fixed phase.

Two independently specifiable phases define a product space. When oscillators with phases $\theta_1(t)$ and $\theta_2(t)$ are considered jointly, each evolving on its own S^1, the state space of the combined system is the product manifold

$$T^2 = S^1 \times S^1, \tag{A.2}$$

which is the two-dimensional torus. This is the space of possible phase pairs; a particular trajectory need not visit all of it. More generally, n independently specifiable circular coordinates define $T^n = (S^1)^n$ [20]. Constraints or locking can restrict the accessible or limiting set.

The torus T^2 can be embedded in three-dimensional Euclidean space via the standard parametric representation:

$$x(\theta, \phi) = (R + r\cos\theta)\cos\phi, \tag{A.3}$$

$$y(\theta, \phi) = (R + r\cos\theta)\sin\phi, \tag{A.4}$$

$$z(\theta, \phi) = r\sin\theta, \tag{A.5}$$

For $R > r > 0$, this is an embedding of the abstract torus, not an inferred neural geometry. Here R is the major radius, defined as the distance from the center of the torus to the center of the tube. The minor radius r is the radius of the tube itself. The angles $\theta, \phi \in [0, 2\pi)$ correspond to the two independent phase variables.

A.3 Coupled Oscillator Dynamics on the Torus

Let $\phi_1, \phi_2 \in \mathbb{R}/2\pi\mathbb{Z}$. For C^1 coupling functions periodic in each phase, the equations

$$\dot{\phi}_1 = \omega_1 + \epsilon f_1(\phi_1, \phi_2), \tag{A.6}$$

$$\dot{\phi}_2 = \omega_2 + \epsilon f_2(\phi_1, \phi_2) \tag{A.7}$$

define a flow on T^2. They do not specify directions transverse to T^2 in a larger state space. Consequently, they cannot establish normal attraction of a neural manifold.

For $\epsilon = 0$ and nonzero frequencies, a rational ratio ω_1/ω_2 gives closed periodic trajectories; an irrational ratio gives trajectories dense in T^2. Both directions are neutral under phase perturbations. These rationality statements apply to constant independent rotations, not to arbitrary coupling functions or merely to the ratio of natural frequencies in a coupled system. For n independent rotations, density requires rational independence of the frequency vector: $k \cdot \omega \neq 0$ for every nonzero $k \in \mathbb{Z}^n$.

A.4 Locking Branches and the Critical Threshold

Subtracting the two equations in Eq. (1) gives Eq. (2). Adding them gives $\dot{\psi} = (\omega_1+\omega_2)/2$ for the locally lifted phase $\psi = (\phi_1 + \phi_2)/2$. The average is used on a lift because it is not a globally single-valued circular coordinate.

For $\epsilon > 0$ and $|\Delta\omega| < 2\epsilon$, solving $\dot{\delta} = 0$ gives the branch δ_* in Eq. (4) and the second branch $\pi - \delta_*$ modulo 2π. Linearizing gives $\dot{\eta} = -2\epsilon \cos\delta_* \, \eta$ to first order. Since $\cos\delta_* > 0$, its exponent is $-\sqrt{4\epsilon^2 - (\Delta\omega)^2}$. The other branch has the opposite sign and is unstable. The common phase remains neutral. A nonzero common frequency makes the stable circle a periodic orbit; zero common frequency gives a circle of equilibria. At $|\Delta\omega| = 2\epsilon$ the exponent vanishes, so exponential stability does not follow.

At the positive threshold $\Delta\omega = 2\epsilon > 0$, an exact solution with $\delta(0) = 0$ is

$$\delta(t) = 2\arctan\left(\frac{\epsilon t}{1+\epsilon t}\right), \qquad \frac{\pi}{2} - \delta(t) \sim \frac{1}{\epsilon t}. \tag{A.8}$$

It follows by setting $u = \tan(\delta/2)$, so that $\dot{u} = \epsilon(u-1)^2$. This orbit approaches the critical phase difference algebraically from below. Perturbations just above $\pi/2$ initially move away before winding around the circle; the critical equilibrium is not Lyapunov stable. Thus a vanishing linear exponent is not evidence for stable exponential locking.

If $|\Delta\omega| > 2\epsilon$, the phase difference drifts. The mean unwrapped slip rate is

$$\langle\dot{\delta}\rangle = \operatorname{sgn}(\Delta\omega)\sqrt{(\Delta\omega)^2 - 4\epsilon^2}. \tag{A.9}$$

For positive detuning this follows by integrating one cycle, $T_\delta = \int_0^{2\pi} (\Delta\omega - 2\epsilon\sin u)^{-1}\, du = 2\pi/\sqrt{(\Delta\omega)^2 - 4\epsilon^2}$. Drift alone is not a proof of dense quasiperiodic motion: the common and slip rotation rates can be commensurate.

The phase model describes motion on a torus but has no ambient normal directions. Appendix B verifies the normal dynamics added in Eq. (3).

B Normal Attraction and Perturbation Calculations

In a specified ambient state space, an attracting invariant torus M is invariant under the flow and has a neighborhood whose trajectories approach it in distance. This local manifold attraction need not imply that every trajectory explores M, or that M is a minimal or transitive attractor. A sufficient mechanism is contraction in normal directions dominating tangent growth in forward and backward time, with appropriate smoothness and compactness assumptions. Normal hyperbolicity gives persistence of the invariant manifold under sufficiently small autonomous C^1 perturbations of the vector field [14]; no frequency nonresonance assumption is needed for that manifold persistence statement.

Equation (3) extends the phase pair with positive radial coordinates r_1, r_2. In the embedding $(r_1 \cos\phi_1, r_1 \sin\phi_1, r_2 \cos\phi_2, r_2 \sin\phi_2) \in \mathbb{R}^4$, the set $M = \{r_1 = r_2 = 1\}$ is an invariant T^2 and

$$r_i(t) - 1 = (r_i(0) - 1)e^{-\lambda t}. \tag{B.1}$$

For positive radii, the nearest point on each unit circle has the same phase. Hence $d(z, M)^2 = (r_1 - 1)^2 + (r_2 - 1)^2$, which gives Eq. (5). This proves normal attraction in the specified model. Replacing $\lambda > 0$ by $\lambda = 0$ or $\lambda < 0$ leaves the flow on M unchanged but makes the normal directions neutral or locally repelling. For the repelling case, the polar description is used while the radii remain positive. Topology and on manifold trajectories alone therefore cannot determine normal stability.

B.1 Normal Hyperbolicity and Internal Motion

The normal attraction can coexist with phase locking. For independent rotations the tangent exponents are zero. For the sinusoidally coupled pair a sufficient domination bound is $\lambda > 2|\epsilon|$, because the tangent Jacobian has norm at most $2|\epsilon|$. Under locking the same torus remains invariant and normally attracting, but its typical limiting motion is the stable circle within it.

Explicitly, the phase Jacobian is

$$J(\phi) = \epsilon \cos\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \|J(\phi)\|_2 \le 2|\epsilon|. \tag{B.2}$$

The forward and inverse tangent propagators are consequently bounded by $e^{2|\epsilon|t}$ for $t \ge 0$. Normal displacements decay as $e^{-\lambda t}$. The stated strict inequality makes normal contraction dominate both tangent bounds. Attraction in the explicit model holds for every $\lambda > 0$; this stronger sufficient bound supplies normal hyperbolicity.

B.2 A Solvable Perturbation

A controlled perturbation gives a second exactly checkable example. With independent phases, replace Eq. (3) by

$$\dot{r}_i = -\lambda(r_i - 1) + a \sin\phi_i. \tag{B.3}$$

The attracting invariant torus is now the graph

$$r_i^*(\phi_i) = 1 + \frac{a(\lambda \sin\phi_i - \omega_i \cos\phi_i)}{\lambda^2 + \omega_i^2}. \tag{B.4}$$

For $|a| < \min_i \sqrt{\lambda^2 + \omega_i^2}$ the graph has positive radii. Substitution gives $\frac{d}{dt}(r_i - r_i^*(\phi_i)) = -\lambda(r_i - r_i^*(\phi_i))$. The graph displacement therefore contracts exactly. Distance to the graph is bounded above by this displacement at the same phases, proving attraction, although distance to the original unit torus need not vanish. This is a special solvable perturbation, not a numerical proof for arbitrary perturbations.

B.3 Scope of Persistence Results

Persistence of a torus as a normally attracting manifold does not ensure that its internal flow remains quasiperiodic. Generic perturbations can change rotation rates and introduce locking. Results preserving quasiperiodic motion require the additional regularity, arithmetic, nondegeneracy, and parameter assumptions of the particular theorem [15].

Classical Hamiltonian KAM results [21–23] cannot simply be applied to a dissipative neural model. Physical energy dissipation alone establishes neither normal hyperbolicity nor those quasiperiodic persistence hypotheses.

C Data Processing and Provenance

C.1 Recording Inventory and Processing

The repository retains five MAT files associated with the Hafting recordings [1]. The earlier release of the data is available from the NIRD archive [11]. We inspect position timestamps, coordinates, and cell timestamp arrays in each file. The audit records input hashes, cell labels, and matrix dimensions in `outputs/claim_audit/empirical_provenance.json`. Separate files are treated as separate recordings unless acquisition metadata establishes simultaneity.

The corrected activity builder bins one selected recording at a time. It uses complete 20 ms bins and divides spike counts by the bin width to obtain rates. Each empirical command requires a recording name and writes to a recording-specific output directory. We separately reconstruct the earlier pooling rule and compare its matrix entry by entry with the retained legacy matrix. This reconstruction identifies provenance; it is not used as a new biological population analysis.

Spatial rate maps use a 50×50 position grid. Spike counts and estimated occupancy time are smoothed separately with a Gaussian kernel of width two spatial bins. Their ratio is evaluated where smoothed occupancy exceeds 0.01 s. Occupancy is approximated by sample count times the median tracking interval. New map generation assigns spikes to the nearest retained position timestamp. Irregular tracking and long gaps require further treatment before quantitative biological interpretation.

C.2 Topology and Entropy Checks

The topology audit reads retained persistence diagrams and inspects the code that generated them. It counts finite H_1 intervals and records their lifetimes. It also checks whether matching H_2 outputs and reported control implementations are present. We do not rerun or calibrate the full neural PCA, UMAP, and persistent homology pipeline in this audit.

Spatial entropy provides a separate description of the retained rate maps. For finite nonnegative rates r_b with positive total, we use

$$p_b = \frac{r_b}{\sum_j r_j}, \qquad H = -\sum_b p_b \log_2 p_b, \tag{C.1}$$

with $0 \log_2 0 = 0$. This weights spatial bins by rate, rather than by occupancy-weighted spike probability. The normalization must therefore accompany any interpretation of the statistic.

The other entropy routines use different categorical variables. The population routine normalizes activity across neuron labels within each time bin. It measures activity-share entropy conditional on positive total activity. The temporal routine partitions samples into contiguous windows and measures the allocation of activity among those windows. Neither routine estimates a process entropy rate or a memory lifetime. Silent or missing distributions are undefined; summaries report how many bins have defined values.

The annotated draft supplied for revision contains additional raster figures and numerical reports. We inspect those reports as part of the document record. Their source recordings, executable notebooks, and numerical arrays are not included in the materials available for this audit. Printed plots alone do not supply a reproducible reconstruction.

C.3 Detailed Audit Results

C.3.1 Recording Provenance

The five retained files contain 23 cell recordings and 13 distinct labels (Table 1). Some labels repeat between trials. These counts do not establish the number of distinct biological cells or a simultaneous pooled population.

Table 1: Retained recording files. Position counts are before removal of nonfinite coordinates. Distinct cell labels across files need not identify distinct biological cells.

File	Position samples	Cell recordings
`Hafting_Fig2c_Trial1.mat`	59,596	7
`Hafting_Fig2c_Trial2.mat`	59,723	7
`Hafting_Fig2d_Trial1.mat`	59,596	3
`Hafting_Fig2d_Trial2.mat`	59,723	3
`rat_10925.mat`	89,652	3

The legacy population matrix does not preserve recording boundaries. Its shape is $(59{,}992, 13)$, and the audit reproduces it exactly by merging matching cell labels across files on one selected recording's time axis. Consequently, its embeddings and population statistics cannot establish simultaneous neural dynamics. The corrected selection rule prevents this pooling.

The retained rate maps describe spatial tuning within the limits of their provenance. Earlier exports reused filenames for repeated cell labels, leaving 13 maps without a one-to-one record of all 23 cell recordings. Those maps do not reconstruct temporal population trajectories. Their geometry cannot establish normal attraction.

C.3.2 Persistent Homology and Control Reports

The retained legacy diagrams provide no calibrated toroidal population result. They contain 32 finite H_1 intervals for PCA and 84 for UMAP. The maximum lifetimes are 0.668947 and 4.001927 in the respective embedding metrics. These values describe stored calculations on the invalid pooled matrix. UMAP changes distances, so the two maxima

are not directly comparable. No matching H_2 output is retained, and the topology entry point computes only H_0 and H_1.

The annotated draft documents further analyses but does not resolve their reproducibility. It displays rate maps, persistence barcodes, and a UMAP embedding attributed to Gardner modules. It also reports an H_2 calculation and a separate Colab parameter sweep. Some numerical values differ from the repository manuscript. We retain these as unverified draft reports, rather than selecting one version as a measured result. In particular, neither a plotted persistence gap nor a stated threshold establishes a noise floor without calibration.

The retained ablation code does not implement the draft's reported timestamp-jitter experiments. It compares spatial smoothing, neuron subsets, and shuffling of binned activity columns. Its legacy outputs inherit the pooling error. Reordering a list of spike timestamps before histogramming leaves the event train unchanged; a temporal surrogate must alter event times or their relationships. Spatial smoothing, temporal smoothing, and PCA whitening also modify different quantities. Their effects cannot be interchanged in an interpretation.

C.3.3 Spatial Entropy

The retained spatial maps have mean entropy 10.860 bits under Eq. (C.1). A uniform distribution over all 2,500 bins has entropy $\log_2(2500) \approx 11.288$ bits. The earlier reported value of approximately 1.49 bits is not reproduced from these maps. Binning, smoothing, missing positions, and normalization affect the comparison.

The stored mean population entropy of 0.260240 bits belongs to the pooled matrix and has a different definition. It is not a valid simultaneous population statistic. Neither entropy value measures temporal information persistence or thermodynamic production. Appendix E examines those separate questions using explicit dynamical and observation models.

C.4 Sampling and Topological Calibration

Temporal sampling must match the frequency being tested. A 100 ms bin gives a sampling rate of 10 Hz and a Nyquist frequency of 5 Hz. It cannot resolve the full 4–8 Hz theta band. Gardner et al. used 10 ms spike count bins with separate smoothing and subsampling choices [2]. Sampling rate, filtering, and the target observable must be specified together.

Topological calibration is a separate requirement. The same neural preprocessing and filtration pipeline should be tested against known topology with matched noise and sampling. Coverage, dependence, metric changes, and preprocessing can alter persistence. Dense sampling alone does not universally inflate spurious H_1 features. The finite oscillator calculations test dynamical equations and do not provide this topology calibration.

D Numerical Methods and Verification

D.1 Simulation Design

All simulations use arbitrary model units. No parameters are fitted to neural recordings. The script `scripts/oscillator_validation.py` uses $(\omega_1, \omega_2) = (1, \sqrt{2})$ and compares $\epsilon = 0$, 0.1 (drift), and 0.6 (locking), with a rational control $(1, 3/2)$ at $\epsilon = 0$. Initial phases are $(0.2, 0.9)$ and the fixed step fourth order Runge–Kutta (RK4) runs use $h = 0.02$ over 200 arbitrary time units. Radial runs use $\lambda = 2$, initial radii $(0.7, 1.3)$, and forcing $a = 0$ or 0.1 over 20 time units. The unforced radial case is also run under phase locking. A separate short control reverses the normal rate without changing the flow on the torus.

D.2 Analytic Benchmarks

The independent runs agree with the exact rotation solution. The rational control returns after 4π. Density for the irrational case follows analytically from the assumed exact frequency ratio; finite samples and floating point arithmetic do not prove it. A 0.2 radian phase kick persists under independent rotation. Locking restores the relative phase but retains a common phase offset.

The coupled runs agree with the locking and slip predictions. At $\epsilon = 0.6$, Eq. (4) gives $\delta_* \approx 0.3524$ rad and $\lambda_\delta \approx -1.1262$ per model time unit. At $\epsilon = 0.1$, Eq. (A.9) gives a mean slip rate of 0.3627. Full cycle crossing times and independent quadrature check the slip rate without endpoint averaging bias. The radial runs agree with the exact contraction and forced-graph solutions in Eqs. (3) and (B.4). Figure 2 compares these behaviors.

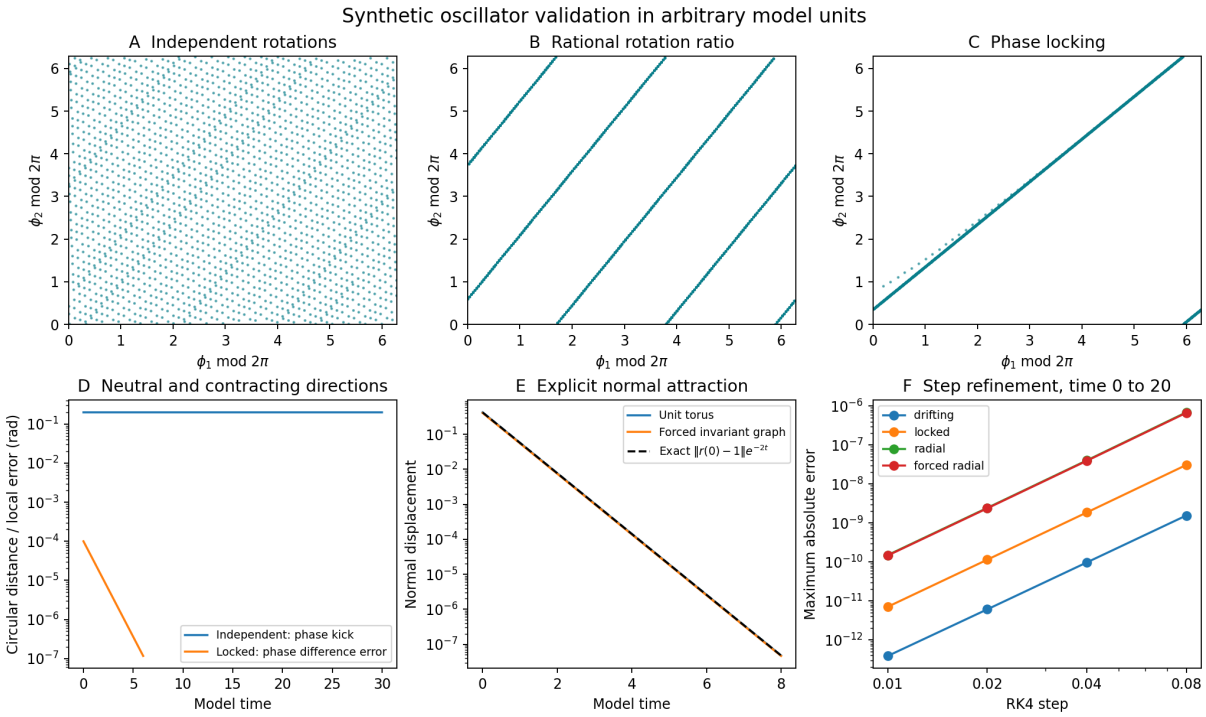


Figure 2: Finite synthetic oscillator calculations in arbitrary units. Phase plots use wrapped coordinates for display; integration uses unwrapped angles. Independent irrational rotations, rational rotations, and phase locking have the same phase coordinate space but different orbit structure. Perturbation checks distinguish neutral phase directions from contraction of phase difference and normal radial displacement. Step refinement checks numerical error against an adaptive reference. None of these panels is a neural recording or empirical attractor reconstruction.

D.3 Numerical Error and Limits

Step refinement measures numerical error over a fixed 20 unit interval. RK4 uses $h = 0.08, 0.04, 0.02, 0.01$. The adaptive DOP853 reference uses relative tolerance 10^{-12}, absolute tolerance 10^{-13}, and maximum step 0.1. A tighter DOP853 run and an independent RK45 integration check the locked case. The errors decrease at approximately fourth order before the numerical floor. At $h = 0.01$, maximum absolute errors against the

reference are approximately 3.9×10^{-13} for drift, 7.1×10^{-12} for locking, and 1.5×10^{-10} for both radial cases. Tests also check exact solutions, stable and unstable branches, and neutral common phase. The CSV trajectories, convergence table, near threshold comparisons, environment versions, and JSON summary are retained with the script.

The supplementary script `scripts/information_validation.py` checks the critical solution in Eq. (A.8) against adaptive integration and the observation examples in Appendix E.2. These calculations also use synthetic parameters and do not estimate biological memory times.

The critical solution is evaluated through $u/(1+u)$, with $u = \epsilon t$, to avoid cancellation near $t = 0$. Quantized bin labels must remain finite and below 2^{53}, where consecutive integers are representable in double precision. Requests beyond that range are rejected rather than interpreted as loss of information. Floating point rounding near bin boundaries and underflow at very long times remain numerical limits of the observation calculation.

Finite agreement checks the implementation for the stated models and intervals. Near the locking threshold, slow drift or recovery can be misclassified over short windows.

Fixed RK4 is conditionally stable: for the scalar normal mode its stability polynomial must satisfy $|1 - z + z^2/2 - z^3/6 + z^4/24| < 1$ with $z = \lambda h$; the tested maximum $z = 0.16$ lies within this interval. These checks do not establish accuracy for arbitrary steps or long time orbit coverage. Stochastic robustness and a neural noise floor are not tested.

E Observation Maps and Entropy Calculations

E.1 Observation Classes and Recursive Processing

An observation map specifies which distinctions a representation retains. Let $\pi : X \to O$ map system states to observations. Define $x \sim_\pi y$ when $\pi(x) = \pi(y)$. Equality makes this relation reflexive, symmetric, and transitive. Its class at x is $[x]_\pi = \{y : \pi(y) = \pi(x)\}$. A noninjective map identifies distinct states; finite dimension alone does not imply noninjectivity.

A deterministic downstream map cannot distinguish states already identified by π. If $R : O \to O$ and $\pi(x) = \pi(y)$, then $R^n(\pi(x)) = R^n(\pi(y))$ for every integer $n \geq 0$. This is a statement about processing the same observation. It does not prohibit new observations or external inputs from providing additional information.

Toroidal coordinates are one possible observation map. Given $\varphi_1, \varphi_2 : X \to S^1$, define $\Phi(x) = (\varphi_1(x), \varphi_2(x)) \in T^2$. Equality of both phase coordinates is equivalent to equality of the ordered pair. This product property alone gives no surjectivity, embedding, or attractor theorem.

E.2 Contraction and Information Persistence

Normal attraction controls distance to a manifold, not the recoverability of an encoded variable. Phase locking suppresses relative phase differences asymptotically, but uniqueness of a smooth ordinary differential equation prevents distinct exact states from merging at finite time on their common interval of existence. This mathematical injectivity need not survive measurement noise, finite resolution, an incomplete observation, or numerical rounding.

For example, encode an equiprobable binary label B in a radial displacement $x_B(0) \in \{0.25, 0.75\}$ and let $x_B(t) = e^{-2t} x_B(0)$. Exact observation distinguishes the labels at every finite time, so $I(B; x_B(t)) = 1$ bit. With the explicitly chosen observation $Q_\Delta(x) = \lfloor x/\Delta \rfloor$, both labels occupy bin zero once $t > \frac{1}{2} \log(0.75/\Delta)$. Their observed mutual information is then zero. At the threshold itself the larger displacement is exactly Δ, which the floor rule assigns to bin one. Halving Δ delays entry into the shared zero bin by $(\log 2)/2$. For $\Delta = 0.1$ and 0.05, these times are approximately 1.0075 and 1.3540 model units. These are consequences of the chosen code and quantizer, not neural memory lifetimes or universal information decay rates.

Differential entropy describes yet another quantity. For a two dimensional normal displacement $X(t) = e^{-\lambda t} X(0)$ with a density and finite differential entropy, the change-

of-variables formula gives

$$h(X(t)) = h(X(0)) + \log|\det(e^{-\lambda t} I_2)| = h(X(0)) - 2\lambda t, \tag{E.1}$$

in nats relative to fixed normal coordinates. This decreasing differential entropy coexists with finite-time invertibility. It is not thermodynamic entropy production; no heat flux follows without a physical model.

E.3 Consistency Across Scales

For a relation between fine and coarse descriptions, one could seek a continuous surjection π and flows F_t, G_t satisfying Eq. (6) for a specified time conversion $c > 0$. Such an autonomous coarse flow requires preservation of observation classes: $\pi(x) = \pi(y)$ must imply $\pi(F_t(x)) = \pi(F_t(y))$. For differentiable maps and vector fields f, g, differentiating at $t = 0$ gives the necessary condition $D\pi_x f(x) = c\, g(\pi(x))$. Establishing a continuous or smooth quotient flow also requires the corresponding topological or regularity assumptions.

The sine pair supplies a direct counterexample to automatic closure. With $\pi(\phi_1, \phi_2) = \phi_1 \pmod{2\pi}$ and $\epsilon \neq 0$, the states $(0, 0)$ and $(0, \pi/2)$ have identical observations but projected velocities ω_1 and $\omega_1 + \epsilon$. Thus this observation does not define an autonomous one phase flow. In contrast, $\sigma = (\phi_1 + \phi_2) \pmod{2\pi}$ satisfies $\dot{\sigma} = \omega_1 + \omega_2$ exactly. The single-phase example also distinguishes repeated processing from a new measurement. For the two fine states just compared, the difference of their lifted first-phase observations at a short time $h > 0$ is $\epsilon h + O(h^2)$ (second state minus first). A later observation can thus separate these states even though any deterministic processing of their identical initial observation gives the same result. The supplementary script checks the two projected velocities and the rotation of the circular sum. Neither these mathematical factors nor the grid cell maps establish continuity or information conservation across molecular, neural, and behavioral scales.

E.4 Entropy and Physical Cost

Thermodynamic entropy requires a physical system boundary and specified exchanges. In units of k_B, its balance is [24]

$$\dot{S} = \dot{S}_i + \dot{S}_e, \qquad \dot{S}_i \geq 0. \tag{E.2}$$

At steady state, $\dot{S}_e = -\dot{S}_i \leq 0$. Export is strictly positive in magnitude only when internal production is positive. This balance does not determine the dimension or topology of a neural manifold.

A proposed entropy funnel must specify the quantity it changes. A restriction of represented states can lower a chosen description length without determining heat export, Shannon entropy, or an entropy rate. Minimum Description Length and predictive coding models motivate conditional compression hypotheses [16–18]. They do not establish decreasing description length at every biological layer. Comparing the metabolic costs of different manifolds remains an open modeling and measurement problem.

F Quaternion Operators and Phase Coordinates

Three independently specifiable circular phases have coordinate space T^3. Unit quaternions instead form S^3, the double cover of $SO(3)$. No continuous embedding of T^3 into S^3 exists. Invariance of domain would make its image open, and compactness would make the image closed. Since S^3 is connected, such an embedding would be a homeomorphism. This contradicts $\pi_1(T^3) = \mathbb{Z}^3$ and $\pi_1(S^3) = 0$ [25].

A separately specified quaternion curve can describe rotation operators. For a pure quaternion angular velocity $\Omega(t)$,

$$\dot{q}(t) = \tfrac{1}{2}\Omega(t)q(t), \qquad |q(0)| = 1 \tag{F.1}$$

preserves $|q(t)| = 1$. This equation does not encode all phase triples or derive their dynamics from three natural frequencies. An action on phase states must be defined separately.

Ordinary phase translations commute. With $T_a(\phi) = \phi + a \pmod{2\pi}$, one has $T_a T_b = T_b T_a$. A noncommutative neural model therefore requires additional operators and a specified composition rule. Comparing $A(B(x))$ and $B(A(x))$ from matched states could test such a rule. Merely changing which frequency band is sampled first also changes its timing and can produce ordinary order dependence.

Pairwise coupling statistics do not specify this operator law. Phase amplitude coupling assigns different roles to a selected phase and amplitude [26]. The weighted phase lag index measures pairwise synchronization through the imaginary cross-spectrum [27]. Neither alone establishes quaternion composition. An empirical test would need controls for delays, common input, amplitudes, and omitted state variables.

G Physical Assumptions Across Frequency Scales

Electromagnetic descriptions require assumptions about sources and material response. In a vacuum region without charges or currents, Maxwell's equations imply [28]

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \partial_t^2 \mathbf{E} = 0. \tag{G.1}$$

For a homogeneous, isotropic, nondispersive Ohmic medium with constant μ, ϵ, σ, no free charge density, and no impressed current, the equation becomes

$$\nabla^2 \mathbf{E} - \mu \epsilon \partial_t^2 \mathbf{E} - \mu \sigma \partial_t \mathbf{E} = 0. \tag{G.2}$$

Neural tissue additionally requires source currents, boundary conditions, and appropriate dispersive properties. These equations do not supply a measured mechanism for toroidal neural dynamics.

Frequency comparisons do not establish coupling between scales. Relative to 10^3 Hz, the range 10^9–10^{13} Hz spans six to ten orders of magnitude. Channel gating times and inverse relaxation times are not automatically coherent oscillation frequencies. A model linking these scales would need measured variables and specified interactions.

The band 0.1–10 THz corresponds to photon energies 0.414–41.36 meV through $E = h\nu$. At 310 K, $k_B T \approx 26.7$ meV [29]. Part of that band is near a thermal energy scale, but the comparison does not establish coherence or equate all photon energies with hydrogen bond dissociation. Molecular spectroscopy and Fröhlich's coherence proposal motivate further questions [30, 31]. They do not demonstrate coupling to the grid cell model in this paper.

H Formalization and Reproducibility

The mathematical checks are analytic calculations and numerical tests, not completed machine proofs. The retained Lean sources contain proof placeholders, undefined identifiers, and no pinned build. Declaring an abstract type named S^1 does not construct the geometric circle. The original prototypes remain in the repository for future development; the paper does not treat them as verification of topology, stability, or empirical evidence.

The repository contains the data audit, simulation scripts, tests, and dependency versions. Run `scripts/oscillator_validation.py` and `scripts/information_validation.py` for the synthetic calculations. Run `scripts/audit_empirical_provenance.py` for the retained-artifact inventory. The README gives the test command, manuscript build instructions, and recording-selection requirements. Repository address: `https://github.com/ezernackchristopher97-cloud/ENTPTCCONPAPER26`.

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